

Review of Last Lecture

7.5 Basic Properties of the z-Transform

**7.6 Relationship Between the z-Transform
and the Laplace Transform**

7.5 Basic Properties of the z-Transform

7.5.1 Linearity

if $ZT[x(n)] = X(z)$ ($R_{x1} < |z| < R_{x2}$), $ZT[y(n)] = Y(z)$ ($R_{y1} < |z| < R_{y2}$)

then $ZT[ax(n) + by(n)] = aX(z) + bY(z)$

$$\max(R_{x1}, R_{y1}) < |z| < \min(R_{x2}, R_{y2})$$

the ROC is the overlapping part

Note: If **some zeros and poles cancel each other out** in the linear combination, **the ROC may expand**.

7.5.2 Shift Property

1. Bilateral z-transform of a Shifted Bilateral Sequence

$$ZT[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$ZT[x(n-m)] = z^{-m}X(z)$$

$$ZT[x(n+m)] = z^mX(z) \quad \mathbf{m} \text{ is any positive integer.}$$

7.5 Basic Properties of the z-Transform

2. Unilateral z-transform of a Left-Shifted Bilateral Sequence

$$ZT[x(n+m)u(n)] = \sum_{n=0}^{\infty} x(n+m)z^{-n} = z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

removing the contribution
of some points

3. Unilateral z-Transform of a Right-Shifted Bilateral Sequence

$$ZT[x(n-m)u(n)] = z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k} \right]$$

adding the contribution
of some points

4. Unilateral z-Transform of a Shifted Causal Sequence $x(n)$

$$ZT[x(n-m)u(n)] = z^{-m} X(z)$$

$$ZT[x(n+m)u(n)] = z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

Sequence shifting only changes the zero-pole situation of the z-transform at **$z=0$ and ∞** .

7.5 Basic Properties of the z-Transform

7.5.3 z-Domain Differentiation (Time-Domain Linear Weighting)

$$\text{if } ZT[x(n)] = X(z)$$

$$ZT[nx(n)] = -z \frac{d}{dz} X(z)$$

Multiplying the time-domain sequence by n is equivalent to differentiating in the z-domain and multiplying by $(-z)$.

$$ZT[n^2 x(n)] = -z \frac{d}{dz} \left[-z \frac{dX(z)}{dz} \right] = \left[-z \frac{d}{dz} \right]^2 X(z)$$

$$ZT[n^m x(n)] = \left[-z \frac{d}{dz} \right]^m X(z)$$

7.5 Basic Properties of the z-Transform

7.5.4 z-Domain Scaling (Time-Domain Exponential Weighting)

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[a^n x(n)] = X\left(\frac{z}{a}\right) \quad (R_{x1} < \left|\frac{z}{a}\right| < R_{x2} \Rightarrow |a|R_{x1} < |z| < |a|R_{x2}) \quad |a| > 1$$

$$ZT[a^{-n} x(n)] = X(az) \quad (R_{x1} < |az| < R_{x2} \Rightarrow \frac{R_{x1}}{|a|} < |z| < \frac{R_{x2}}{|a|}) \quad |a| > 1$$

$$ZT[(-1)^n x(n)] = X(-z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[(-1)^n u(n)] = z/(z+1) \quad (|z| > 1)$$

Given $X(z) = \frac{z}{z^2 - \frac{1}{6}z - \frac{1}{6}} \left(|z| > \frac{1}{2} \right), \quad \lim_{n \rightarrow \infty} x(n) = (\quad)$

- ☐ A 1
- ☐ B -1
- ☒ C 0
- ☐ D 2

提交

7.5 Basic Properties of the z-Transform

7.5.5 Initial Value Theorem

If $\mathbf{x(n)}$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $x(0) = \lim_{z \rightarrow \infty} X(z)$ (the signal jump at $\mathbf{n=0}$ corresponds to high-frequency components)

7.5.6 Final Value Theorem

If $\mathbf{x(n)}$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $\lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} [(z-1)X(z)]$ ($\mathbf{n \rightarrow \infty}$ corresponds to the DC component, i.e., $\mathbf{s=0, z=e^s=1}$)

It requires that $\mathbf{x(n)}$ converges as $n \rightarrow \infty$, i.e., the poles of $X(z)$ must lie **inside the unit circle (if on the unit circle, it should be a first-order pole and can only be at $z=1$).**

Note the difference from the system stability condition: the condition **for the stability of a causal system** is that the poles of the system function must **lie inside the unit circle.**

7.5 Basic Properties of the z-Transform

7.5.7 Time-Domain Convolution Theorem

Given two sequences $x(n)$ and $h(n)$, their z-transforms are:

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[h(n)] = H(z) \quad (R_{h1} < |z| < R_{h2})$$

Then the z-transform of the time-domain convolution of the two sequences is equivalent to the product of their z-transforms in the z-domain:

$$ZT[x(n) * h(n)] = X(z)H(z)$$
$$\max(R_{x1}, R_{h1}) < |z| < \min(R_{x2}, R_{h2})$$

The ROC is the overlapping part.

Note: If some zeros and poles cancel each other out, the ROC may expand.

Mapping Relationship Between the z-Plane and s-Plane Translation

$$z = e^{sT} \quad T \text{ is the time interval of the sequence}$$

Express s in rectangular coordinate form and **z in polar coordinate form:**

$$s = \sigma + j\omega, \quad z = re^{j\theta}$$

$$z = e^{(\sigma + j\omega)T} = e^{\sigma T} e^{j\omega T} = re^{j\theta}$$

$$\left\{ \begin{array}{l} r = e^{\sigma T} = e^{\frac{2\pi\sigma}{\omega_s}} \\ \theta = \omega T = 2\pi \frac{\omega}{\omega_s} \end{array} \right.$$

$$\omega_s = \frac{2\pi}{T} \text{ is the repetition frequency}$$

7.6 Relationship Between the z-Transform and the Laplace Transform



1) The imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ the unit circle ($r = 1$) in the z-plane

$$\sigma = 0, s = j\omega \quad r = |z| = e^{\sigma T} = 1$$

2) The left half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the interior of the unit circle ($r < 1$) on the z-plane

$$\sigma < 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} < 1$$

3) The right half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the exterior of the unit circle ($r > 1$) on the z-plane

$$\sigma > 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} > 1$$

4) Lines parallel to the imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ circles on the z-plane $\begin{pmatrix} \sigma > 0, r > 1 \\ \sigma < 0, r < 1 \end{pmatrix}$

7.6 Relationship Between the z-Transform and the Laplace Transform



5) The real axis ($\omega=0$) on the s-plane $\xrightarrow{\text{mapping}}$ the positive real axis ($\theta=0$) on the z-plane

Lines parallel to the real axis (ω is a constant) $\xrightarrow{\text{mapping}}$

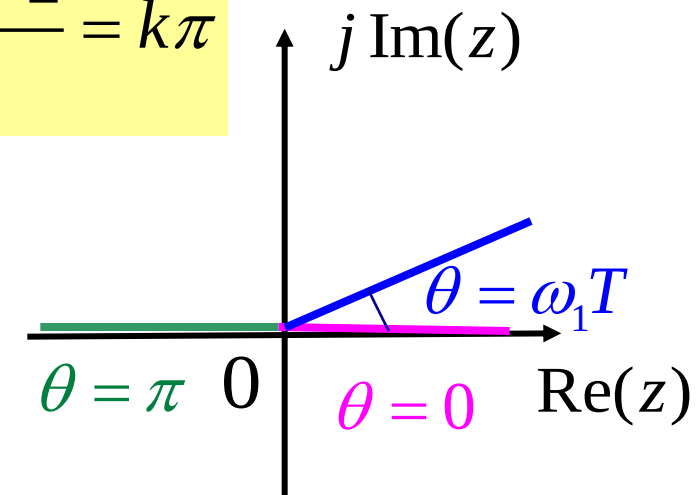
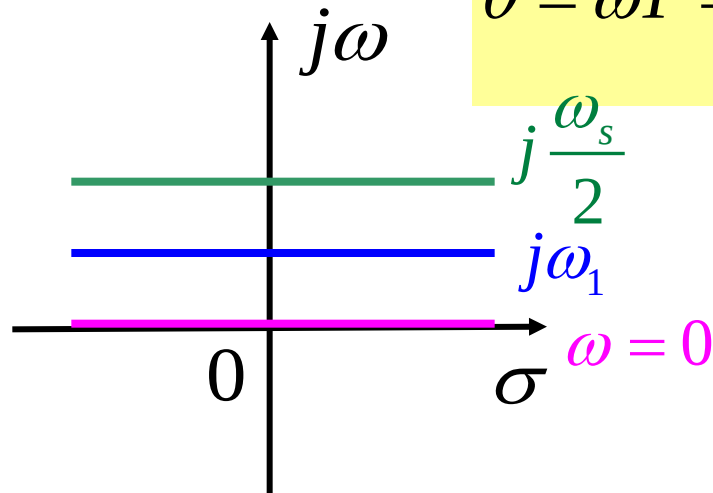
rays originating from the origin on the z-plane

Lines parallel to the real axis (passing through $j \frac{k\omega_s}{2}$ ($k=\pm 1, \pm 3, \dots$)) in the s-plane $\xrightarrow{\text{mapping}}$

the negative real axis in the z-plane $\left(\begin{matrix} \theta = \pi \\ r \text{ is arbitrary} \end{matrix} \right)$

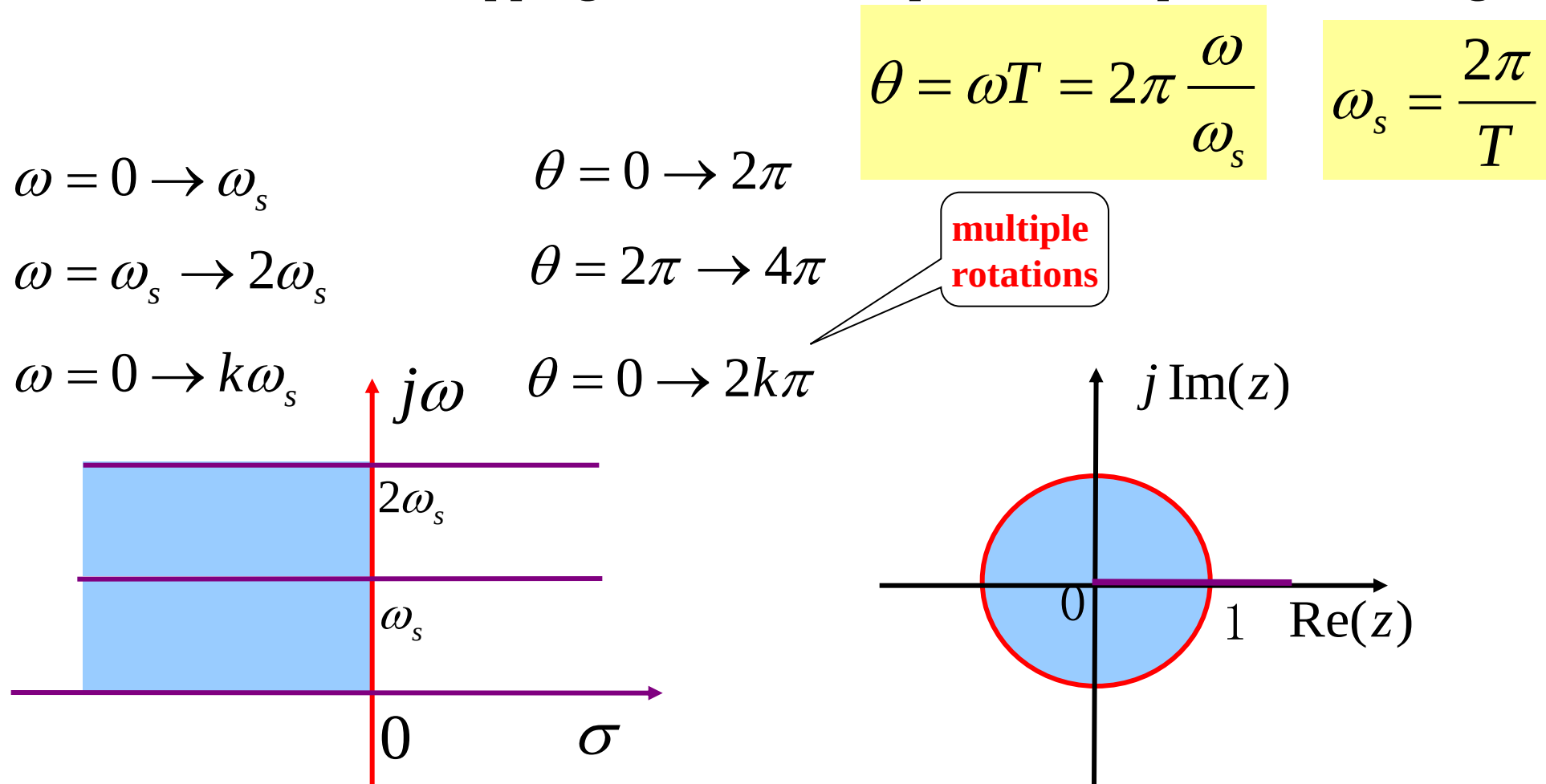
$$\omega_s = \frac{2\pi}{T}$$

$$\theta = \omega T = 2\pi \frac{\omega}{\omega_s} = 2\pi \frac{k\omega_s / 2}{\omega_s} = k\pi$$

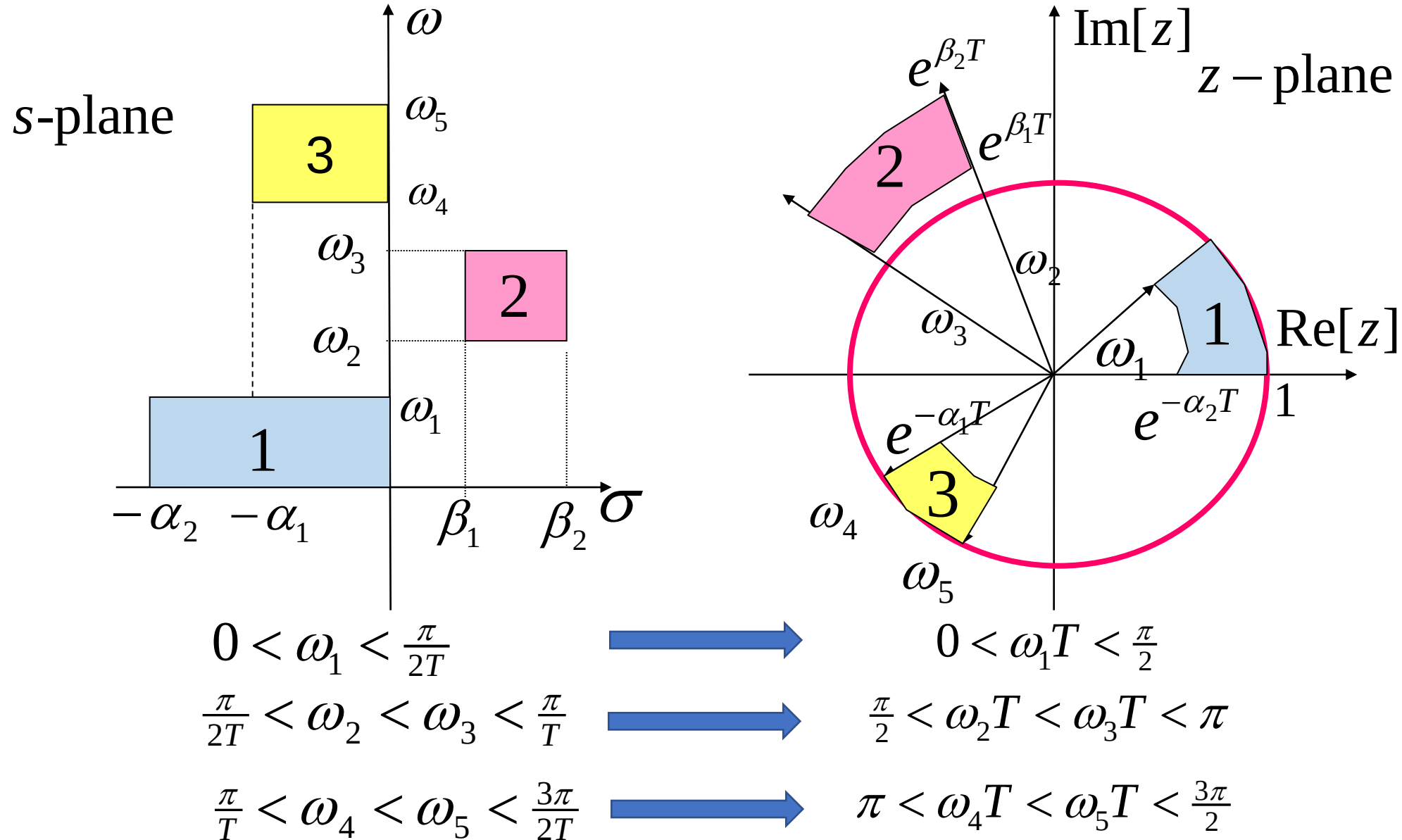


7.6 Relationship Between the z-Transform and the Laplace Transform

6) Moving along the imaginary axis in the s – plane $\xrightarrow{\text{mapping}}$ periodic rotation along the unit circle in the z – plane: each translation by ω_s corresponds to one full rotation around the unit circle, i.e., the mapping between the s-plane and z-plane is not single-valued.



7.6 Relationship Between the z-Transform and the Laplace Transform



Contents of This Lecture

7.7 Solving difference equations
using z-transform

7.8 System function of discrete-time systems

7.9 Signal flow graph of systems

Learning Objectives of This Lecture

1. Proficiency in calculating the **zero-state response** using z-transform;
2. Proficiency in the mapping relationships among **system functions, difference equations, and simulation block diagrams**;
3. Understand the relationship between **the pole distribution of the system function** and the characteristics of **the unit sample response**;
4. Familiarize with the methods for determining system **stability** and **causality**;
5. Given a **differential/difference equation/system function**, be able to proficiently draw the system's signal flow graph.

7.7 Solving Difference Equations using Z-Transform

Principle: Utilizing the **linearity** and **shift** property of the **unilateral z-transform**

1. Shift property of the unilateral z-transform

$$X(z) = \sum_{n=0}^{\infty} x(n)u(n)z^{-n}$$

$$ZT[x(n-m)u(n)] = z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k} \right]$$

$$ZT[x(n-1)u(n)] = z^{-1}X(z) + x(-1)$$

$$ZT[x(n-2)u(n)] = z^{-2}X(z) + z^{-1}x(-1) + x(-2)$$

7.7 Solving Difference Equations using Z-Transform

2. Steps for solving difference equations using the unilateral z-transform

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

Both $x(n-r)$ and $y(n-k)$ are right-shifted sequences.

Take the unilateral z-transform of both sides:

$$\sum_{k=0}^N a_k z^{-k} [Y(z) + \sum_{l=-k}^{-1} y(l) z^{-l}] = \sum_{r=0}^M b_r z^{-r} [X(z) + \sum_{m=-r}^{-1} x(m) z^{-m}]$$

Initial state

If it is a causal signal,
this term is zero

7.7 Solving Difference Equations using Z-Transform

Zero-State Response

If the initial state of the system is $\mathbf{y}(l)=\mathbf{0}$ ($-N \leq l \leq -1$), the system is in the zero state, and the excitation $\mathbf{x}(n)$ is a causal sequence, then:

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{r=0}^M b_r z^{-r} X(z)$$

The result obtained is the zero-state response:

$$Y_{zs}(z) = X(z) \cdot \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

System Function:

$$H(z) = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

$$Y_{zs}(z) = X(z) H(z)$$

$$y_{zs}(n) = ZT^{-1}[X(z)H(z)]$$

7.7 Solving Difference Equations using Z-Transform

Example 7-10: Given the difference equation $y(n) - 0.9y(n-1) = 0.05x(n)$ of a causal system, with the excitation $x(n) = u(n)$ (unit step sequence), find the zero-state response $y_{zs}(n)$ of the system using the inverse z-transform method.

Solution: Take the z-transform of both sides of the equation:

$$Y(z) - 0.9z^{-1}Y(z) = \frac{0.05z}{z-1} \quad \text{Using linearity and shift property}$$

$$Y(z) = \frac{0.05z^2}{(z-0.9)(z-1)} \quad \frac{Y(z)}{z} = \frac{A_1}{z-0.9} + \frac{A_2}{z-1}$$

$$A_1 = \left. \frac{0.05z}{z-1} \right|_{z=0.9} = -0.45 \quad A_2 = \left. \frac{0.05z}{z-0.9} \right|_{z=1} = 0.5$$

$$Y_{zs}(z) = -\frac{0.45z}{z-0.9} + \frac{0.5z}{z-1} \quad (|z| > 1)$$

$$y_{zs}(n) = [-0.45(0.9)^n + 0.5]u(n)$$

7.8 System Function of Discrete-Time Systems

7.8.1 Definition of the System Function

1. Ratio of the z-transform of the system's zero-state response to the z-transform of the excitation

If $x(n)$ is a **causal sequence** and the system is in the **zero state**:

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

Zero-state

$$Y(z) \sum_{k=0}^N a_k z^{-k} = X(z) \sum_{r=0}^M b_r z^{-r}$$

Causal

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

2. The z-transform of the unit sample response $h(n)$

The convolution sum of the excitation and the unit sample response is the zero-state response of the system:

$$y_{zs}(n) = x(n) * h(n)$$

By the time-domain convolution theorem:

$$Y_{zs}(z) = X(z)H(z)$$

$$H(z) = \frac{Y_{zs}(z)}{X(z)}$$

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

What is the unit sample response of the discrete-time system described by the difference equation $y(n) - ay(n-1) = bx(n)$? ()

- ☐ A $h(n) = ab^n u(n)$
- ☐ B $h(n) = \left(\frac{a}{b}\right)^n u(n)$
- ☒ C $h(n) = ba^n u(n)$
- ☐ D $h(n) = \left(\frac{b}{a}\right)^n u(n)$

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7.8 System Function of Discrete-Time Systems

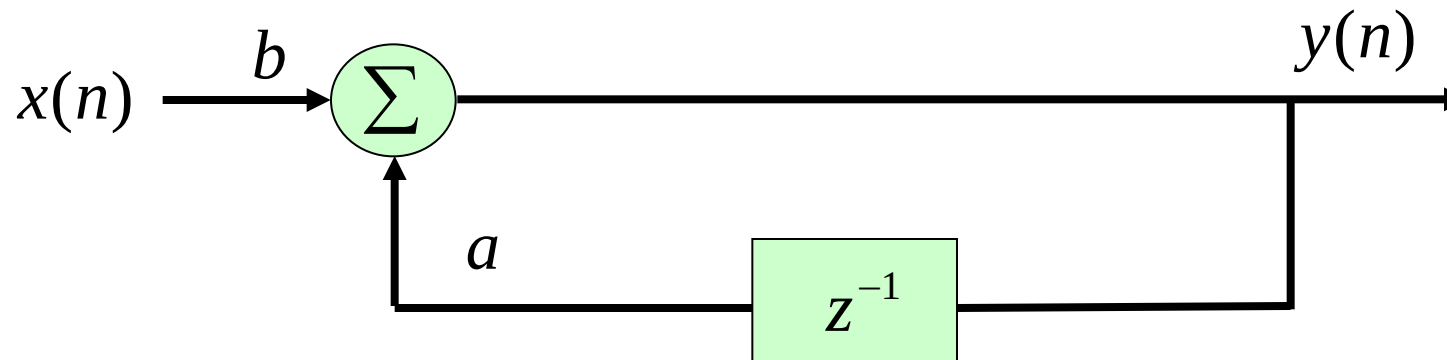
Example 7-11: Find the system function and unit sample response of the discrete-time system described by the following difference equation, and draw the system block diagram.

$$y(n] - ay(n-1) = bx(n]$$

Solution: If the system is in the zero state, then $y(-1)=0$. Taking the z-transform of both sides of the equation gives:

$$Y(z)(1 - az^{-1}) = bX(z)$$

$$H(z) = \frac{b}{1 - az^{-1}} = \frac{bz}{z - a} \quad (|z| > |a|) \quad h(n) = ba^n u(n)$$



7.8 System Function of Discrete-Time Systems

Example 7-12: Suppose the system function of a second-order discrete linear time-invariant system is:

$$H(z) = \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1+\frac{3}{4}z^{-1})}$$

Find the difference equation corresponding to this system, and draw the system simulation block diagram.

Solution:

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$\left(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}\right)Y(z) = (1 + 2z^{-1} + z^{-2})X(z)$$

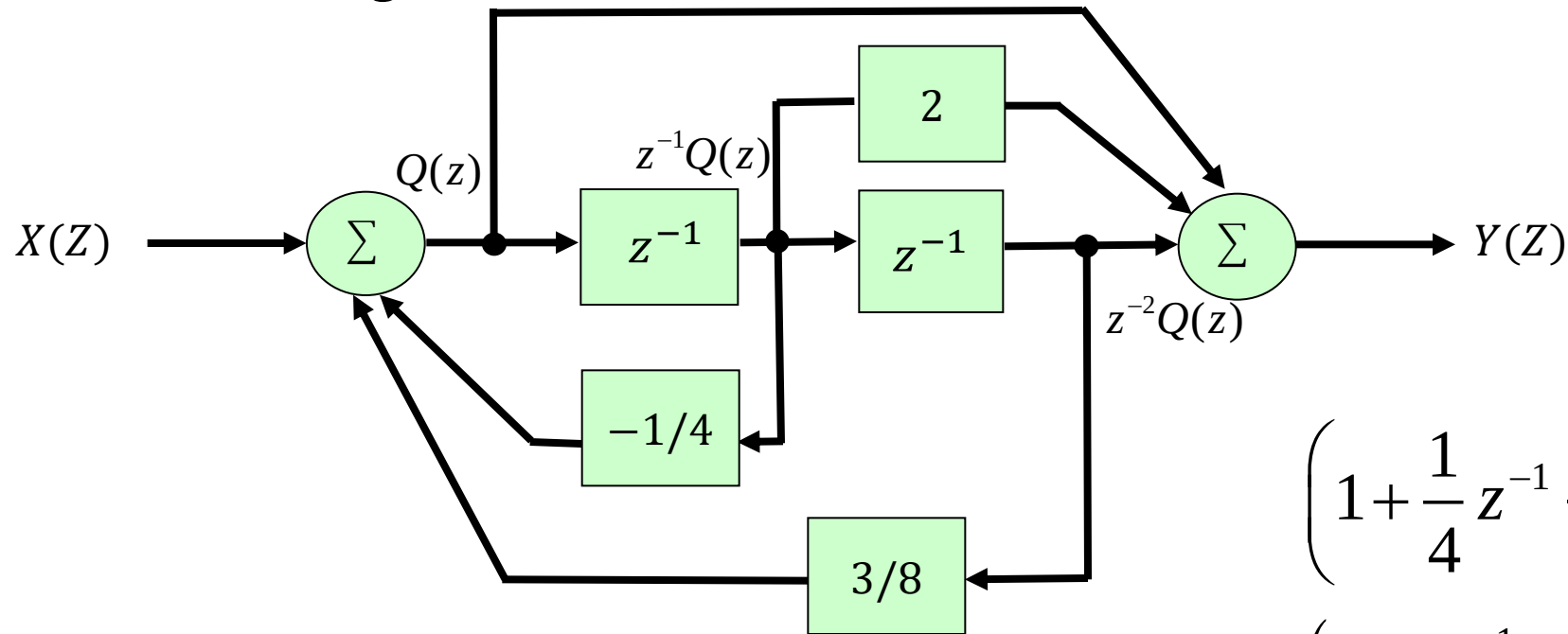
The system difference equation is

$$y(n) + \frac{1}{4}y(n-1) - \frac{3}{8}y(n-2) = x(n) + 2x(n-1) + x(n-2)$$

7.8 System Function of Discrete-Time Systems

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

system block diagram:



$$\left(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}\right)Q(z) = X(z)$$

$$(1 + 2z^{-1} + z^{-2})Q(z) = Y(z)$$

7.8 System Function of Discrete-Time Systems

7.8.2 Analyzing the Unit Sample Response by the Zero-Pole Distribution of the System Function

$$H(z) = \left[G \frac{\prod_{r=1}^M (1 - z_r z^{-1})}{\prod_{k=1}^N (1 - p_k z^{-1})} \right] = \sum_{k=0}^N \frac{A_k z}{z - p_k} = A_0 + \sum_{k=1}^N \frac{A_k z}{z - p_k}$$

where z_r and p_k are the zeros and poles of $H(z)$ respectively; $p_0 = 0$

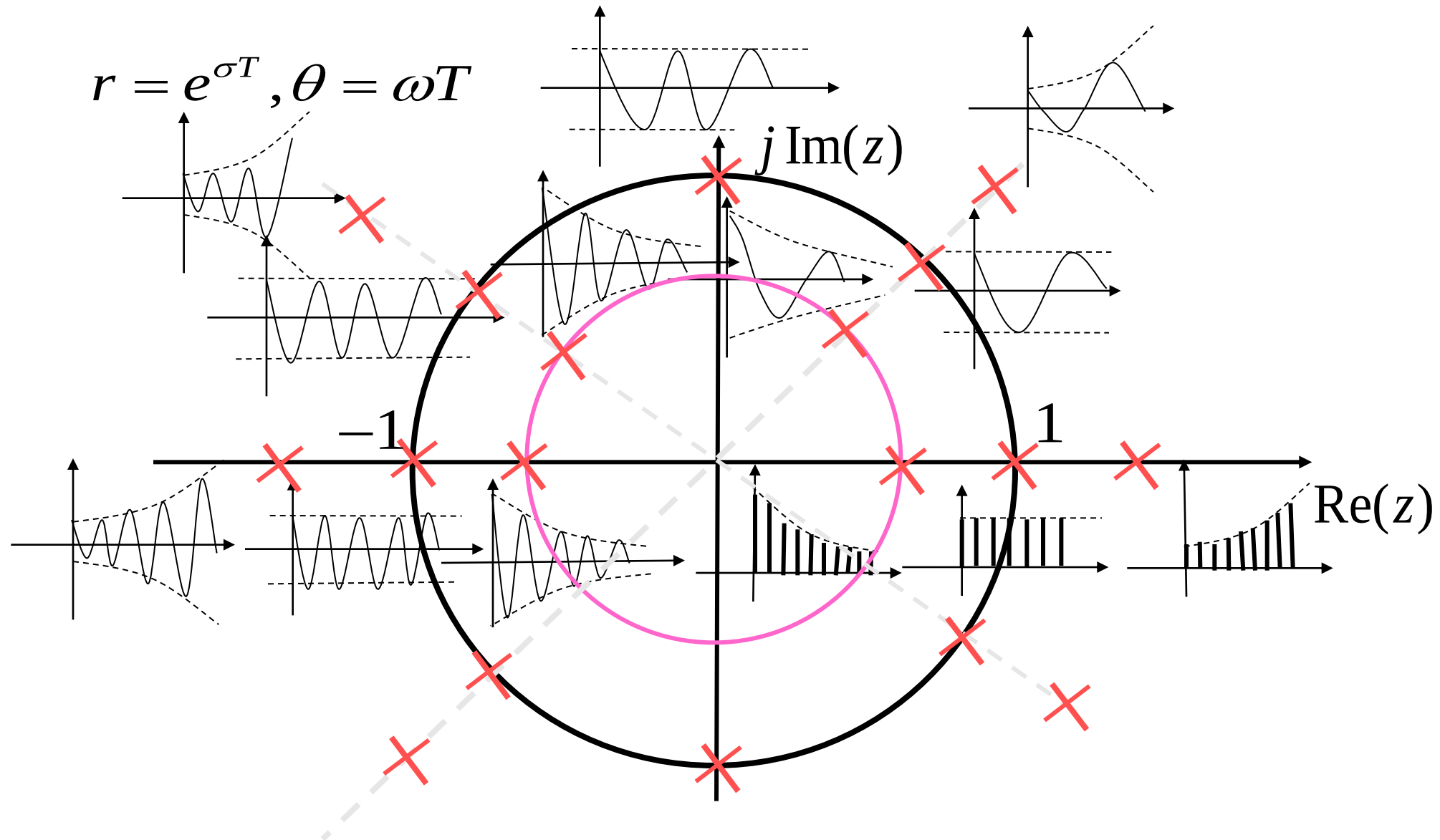
$$h(n) = ZT^{-1}[H(z)] = A_0 \delta(n) + \sum_{k=1}^N A_k (p_k)^n u(n)$$

$\therefore$ The characteristics of $h(n)$ depend on the poles p_k of $H(z)$

The amplitude A_k of $h(n)$ depends on the zeros z_r of $H(z)$

7.8 System Function of Discrete-Time Systems

Effects of the Pole Distribution on $h(n)$



7.8 System Function of Discrete-Time Systems

Example 7-13: Given the difference equation of the system below, find the unit sample response of the system using the z-domain analysis.

$$y(n) - 5y(n-1) + 6y(n-2) = x(n) - 3x(n-2)$$

Solution: (1) Take the z-transform of both sides of the difference equation, we get

$$(1 - 5z^{-1} + 6z^{-2})Y(z) = (1 - 3z^{-2})X(z)$$

(2) The system function is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - 3z^{-2}}{1 - 5z^{-1} + 6z^{-2}} = \frac{z^2 - 3}{z^2 - 5z + 6}$$

$$\frac{H(z)}{z} = \frac{z^2 - 3}{z(z-2)(z-3)} = -\frac{1}{2z} - \frac{\frac{1}{2}}{z-2} + \frac{2}{z-3}$$

$$H(z) = -\frac{1}{2} - \frac{1}{2} \cdot \frac{z}{z-2} + \frac{2z}{z-3} \quad (|z| > 3)$$

(3) Take the inverse z-transform of the system function, the unit sample response is:

$$h(n) = -\frac{1}{2}\delta(n) + (-2^{n-1} + 2 \times 3^n)u(n) = \delta(n) + 5\delta(n-1) + (2 \times 3^n - 2^{n-1})u(n-2)$$

7.8 System Function of Discrete-Time Systems

7.8.3 Stability and Causality of Discrete-Time Systems

1. Conditions for Causal Stability of the System in the Time Domain

The **necessary and sufficient condition** for the stability of a discrete-time system is that the **unit sample response is absolutely summable**, i.e.:

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

The **necessary and sufficient condition** for a **causal stable system** is that **$h(n)$** is a **causal sequence and absolutely summable**, i.e.:

$$\left\{ \begin{array}{ll} h(n) = h(n)u(n) & \text{Causal} \\ \sum_{n=-\infty}^{\infty} |h(n)| < \infty & \text{Stable} \end{array} \right.$$

2. Conditions for Causal Stability in the z-Domain

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n}, \text{ its existence requires: } \sum_{n=-\infty}^{\infty} |h(n)z^{-n}| < \infty$$

when $|z| = 1$, $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$, just satisfies the condition for system stability.

Therefore, for a **stable system**, the ROC of $H(z)$ should **include the unit circle** (regardless of whether the system is causal).

For a **causal stable system**, the ROC is $|z| > a$ ($a < 1$), which means **all poles lie inside the unit circle**.

7.8 System Function of Discrete-Time Systems



Example 7-14: Given the system function below, explain the stability of the system under three ROC conditions, and find the unit sample response.

$$H(z) = \frac{-9.5z}{(z - 0.5)(z - 10)} \quad \begin{array}{ll} (1) 10 < |z| \leq \infty & (2) 0.5 < |z| < 10 \\ (3) |z| < 0.5 & \end{array}$$

Solution: (1) $10 < |z| \leq \infty$

Method 1: From the ROC, this is a causal system. The ROC does not include the unit circle, so the system is unstable.

Method 2: Judge from the ROC that this system is causal.

$$z_1 = 0.5 \quad z_2 = 10 \quad |z_2| > 1$$

The poles of a causal stable system lie inside the unit circle. Here, one pole is outside the unit circle, so the system is unstable.

$$h(n) = [(0.5)^n - (10)^n]u(n)$$

7.8 System Function of Discrete-Time Systems



$$(2) \quad 0.5 < |z| < 10$$

Right-sided
sequence

Left-sided
sequence

The ROC includes the unit circle, so the system is stable.

Judge from the ROC that this system is non-causal.

$$H(z) = \frac{z}{z - 0.5} - \frac{z}{z - 10}$$

$$h(n) = (0.5)^n u(n) + (10)^n u(-n-1)$$

$$(3) \quad |z| < 0.5 \quad \text{Left-sided sequence}$$

The ROC does not include the unit circle, so the system is unstable.

Judge from the ROC that this system is non-causal.

$$h(n) = [(10)^n - (0.5)^n] u(-n-1)$$

From the above, **the stability condition for a non-causal system** is also that **the ROC of the system function includes the unit circle.**

Given the difference equation of a causal system:

$$y(n) + 0.2y(n - 1) - 0.24y(n - 2) = x(n) + x(n - 1)$$

Is this system stable?

- ☒ A Stable
- ☐ B Unstable

提交

7.8 System Function of Discrete-Time Systems

Example 7-15:

Given $y(n) + 0.2y(n-1) - 0.24y(n-2) = x(n) + x(n-1)$, find $H(z)$ and $h(n)$, discuss the ROC and stability of this causal system. If the excitation signal is $x(n) = u(n)$, find the zero – state response.

Solution : (1) Find the system function

Take the z-transform of both sides of the difference equation:

$$Y(z) + 0.2z^{-1}Y(z) - 0.24z^{-2}Y(z) = X(z) + z^{-1}X(z)$$

Rearrange to get:
$$\left(1 + \frac{0.2}{z} - \frac{0.24}{z^2}\right)Y(z) = \left(1 + \frac{1}{z}\right)X(z)$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{z(z+1)}{z^2 + 0.2z - 0.24} = \frac{z(z+1)}{(z-0.4)(z+0.6)}$$

7.8 System Function of Discrete-Time Systems



(2) Find the ROC and determine stability

$$H(z) = \frac{z(z+1)}{(z-0.4)(z+0.6)} \quad \because z_1 = 0.4, \quad z_2 = -0.6$$

The system is causal, $\therefore |z| > 0.6$

The ROC includes the unit circle, so the system is stable.

(3) Find the unit sample response $h(n)$

$$\frac{H(z)}{z} = \frac{1.4}{z-0.4} - \frac{0.4}{z+0.6} \quad H(z) = \frac{1.4z}{z-0.4} - \frac{0.4z}{z+0.6}$$

$$h(n) = [1.4(0.4)^n - 0.4(-0.6)^n]u(n)$$

7.8 System Function of Discrete-Time Systems



(4) Find the zero-state response

$$x(n) = u(n) \qquad X(z) = \frac{z}{z-1} \quad (|z| > 1)$$

$$Y_{zs}(z) = H(z) X(z) = \frac{z^2(z+1)}{(z-1)(z-0.4)(z+0.6)}$$

$$Y_{zs}(z) = \frac{2.08z}{z-1} - \frac{0.93z}{z-0.4} - \frac{0.15z}{z+0.6} \quad (|z| > 1)$$

zero-state response

$$y_{zs}(n) = [2.08 - 0.93(0.4)^n - 0.15(-0.6)^n]u(n)$$

7.9 Signal Flow Graph of Systems

1. Overview

Systems (continuous or discrete) can be described using block diagrams, which are more intuitive than differential equations or difference equations.

Simulation of linear systems:

- Continuous systems — addition, multiplication, integration
- Discrete systems — addition, multiplication, delay

System block diagram $\xrightarrow{\text{simplify}}$ **Signal flow graph**

The **Signal Flow Graph** was proposed by Mason (from MIT) in the 1950s. It is applied in feedback system analysis, linear equation system solving, linear system simulation, digital filter design, etc.

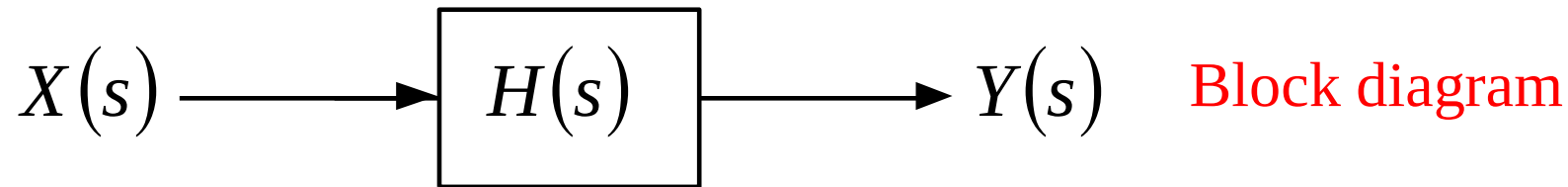
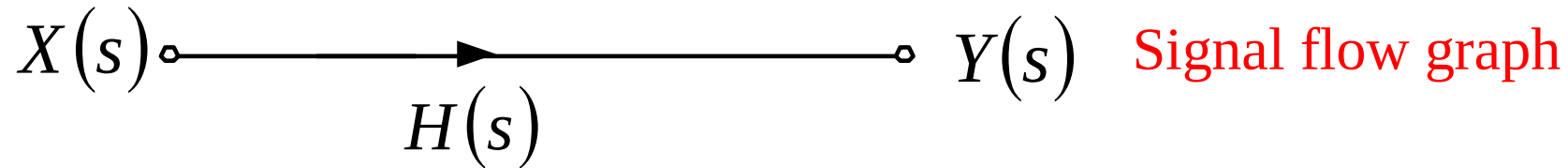
Key advantages of the signal flow graph method:

- The system model is represented concisely and clearly;
- Simplifies the calculation equations of the system function.

7.9 Signal Flow Graph of Systems

2. Representation of Systems Using Signal Flow Graphs

In fact, systems are described using **nodes** and **branches**:



$X(s)$ 、 $Y(s)$ are called **nodes**.

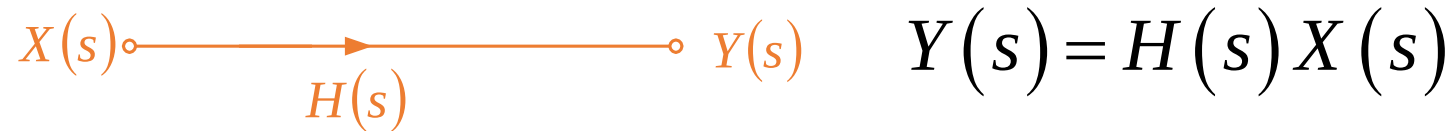
The line segment representing the signal transmission path is called a **branch**.

The signal transmission direction is indicated by an arrow; the **system function (transfer function) is marked near the arrow, which acts as a multiplier.**

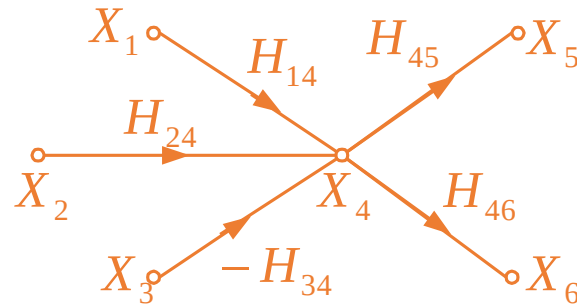
7.9 Signal Flow Graph of Systems

3. Properties of Signal Flow Graphs

(1) A branch represents the functional relationship between two signals; signals can only pass along the direction of the arrow on the branch:



(2) A node can superimpose the signals of all input branches, and transmit the summed signal to all output branches:



For example, node X_4

$$X_4 = H_{14}X_1 + H_{24}X_2 - H_{34}X_3$$

(3) The **transfer function** remains unchanged after the signal flow graph is transposed. Transposition means reversing the signal transmission direction of all branches in the graph, while swapping the input and output nodes.

(4) For a given system, **the signal flow graph is not unique in form.**

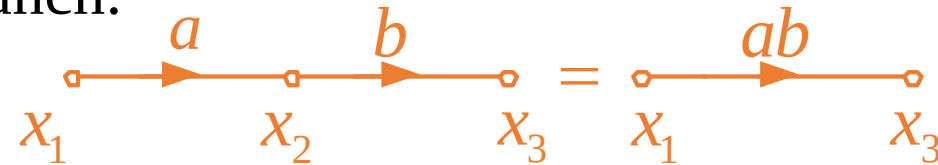
7.9 Signal Flow Graph of Systems

4. Algebraic Operations of Signal Flow Graphs

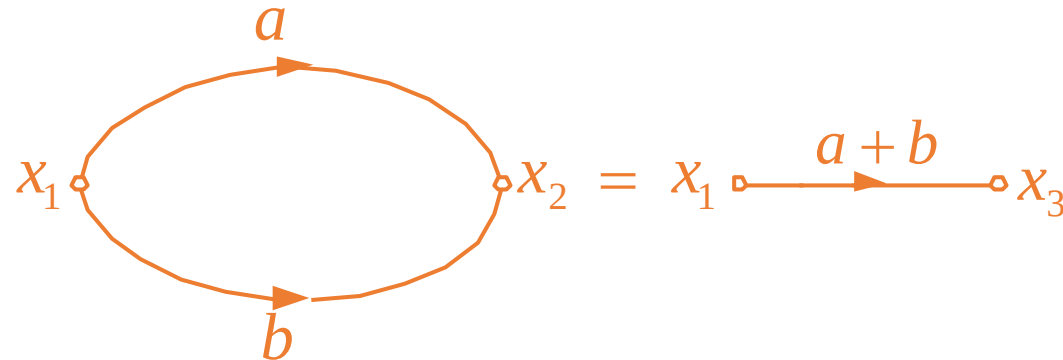
(1) The value of a node with one input branch equals the input signal multiplied by the branch gain:



(2) **Merging of series branches:** The total gain equals the product of the gains of each branch:

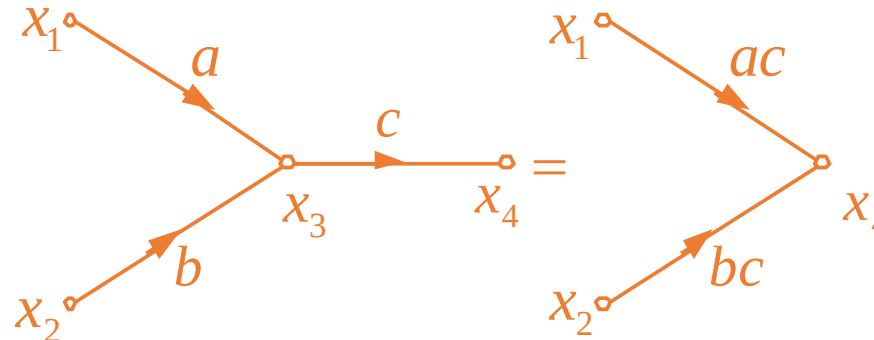


(3) **Merging of parallel branches:** Sum the gains of parallel branches

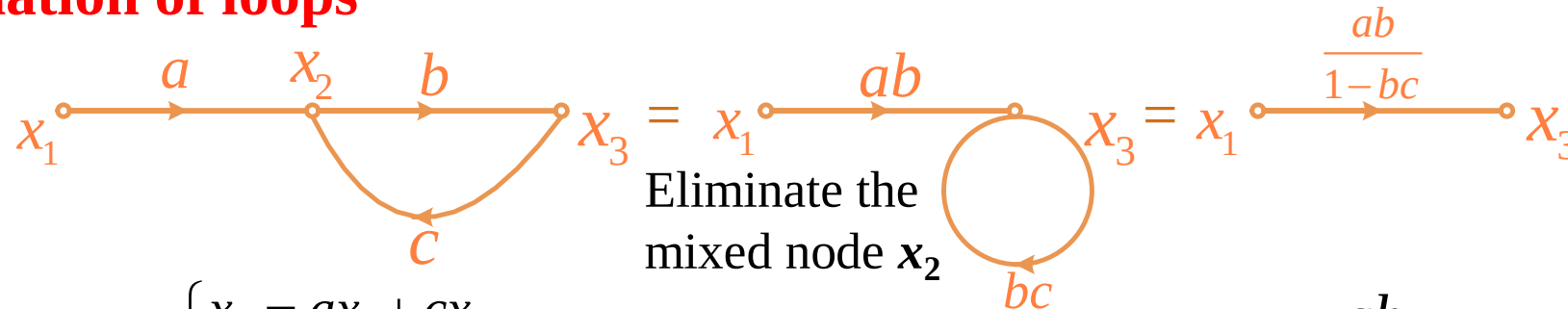


7.9 Signal Flow Graph of Systems

(4) Elimination of mixed nodes



(5) Elimination of loops



$$\text{Thus } \begin{cases} x_2 = ax_1 + cx_3 \\ x_3 = bx_2 \end{cases} \Rightarrow x_3 = abx_1 + bcx_3 \Rightarrow x_3 = \frac{ab}{1-bc} x_1$$

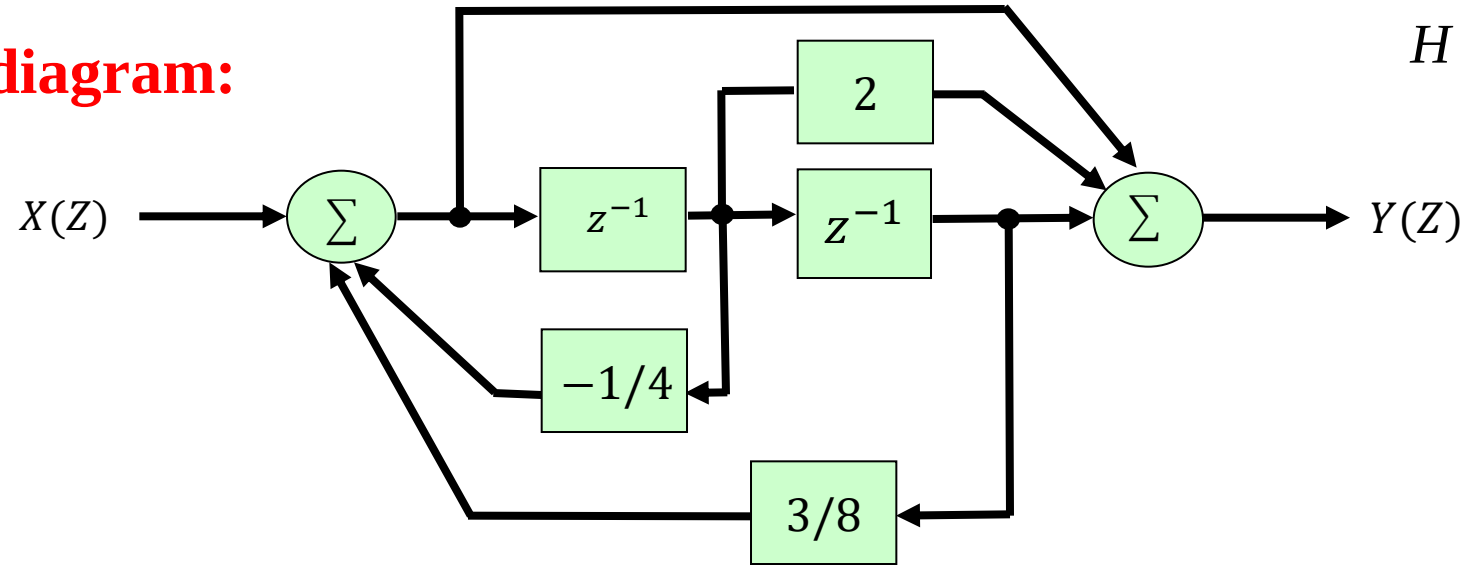
Summary: The signal flow graph can be simplified through the following steps to obtain the system function:

- ① Merge series branches to reduce nodes;
- ② Merge parallel branches to reduce branches;
- ③ Eliminate loops

7.9 Signal Flow Graph of Systems

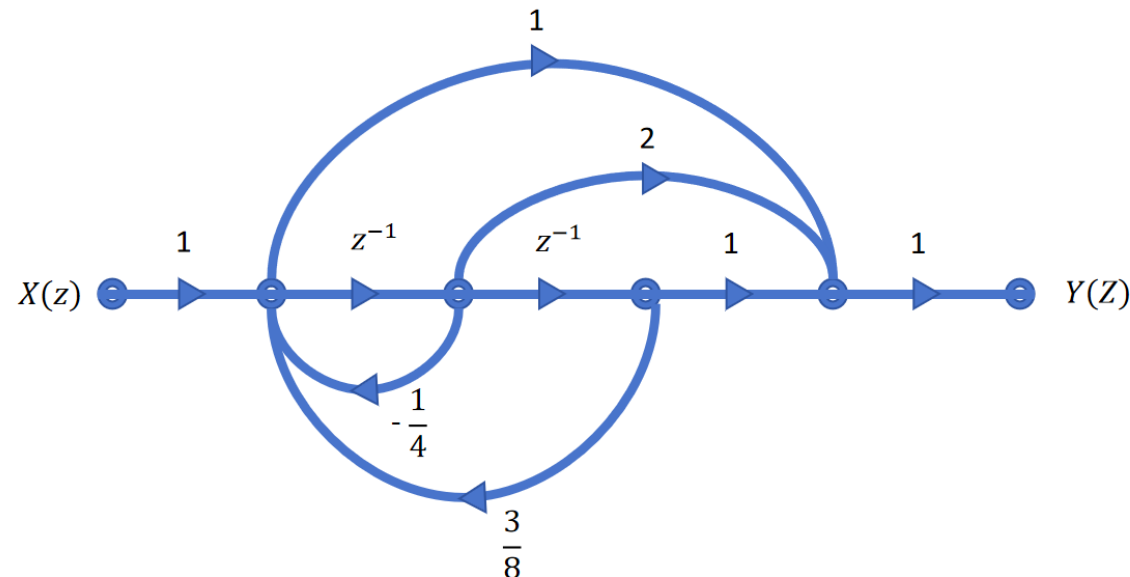
Example 7-12

Block diagram:



$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

Signal flow graph:



1. Repeat all the examples in the lecture notes.
2. 7.34(a) (judge only causality and stability) in the textbook.
3. Given the following system difference equation, find the system function $H(z)$ and the unit sample response $h(n)$. If the excitation signal is $x(n) = u(n)$, find the zero-state response.

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = 2x(n)$$